Customer Shopping Behaviour Analysis

End-to-end project: Python > PostgreSQL > Power BI

Data cleaning, SQL exploratory analysis, interactive dashboard and business insights

Executive summary

What 3,900 customer transactions tell us

3,900	$233,081	$59.76	3.75 / 5
Customers analysed	Total revenue	Average purchase	Average review rating

Headline findings

- Spending is remarkably uniform: gender, subscription, shipping and discount groups all average $58 to $61 per purchase, so revenue is driven by customer volume, not basket size.
- Males generate 68% of revenue ($157.9K) simply because they are 68% of customers; Clothing alone brings in 45% of revenue.
- 80% of customers are already Loyal (more than 10 prior purchases), yet only 27% subscribe, which is the clearest growth lever.
- Discounts reach 43% of purchases without lifting basket size ($59.28 with discount vs $60.13 without).

Project workflow

Four stages from raw CSV to decision-ready dashboard

1

Raw data

customer_shopping_behavior.csv: 3,900 rows, 18 columns

>

2

Python

Jupyter notebook: clean, transform, engineer features

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3

PostgreSQL

Loaded into table "customer"; 10 EDA queries

>

4

Power BI

Connected to SQL; interactive KPI dashboard

Goal: understand who buys, what they buy, how they pay and ship, and which levers (discounts, subscriptions) move revenue.

Dataset and Python preparation

Jupyter notebook turned the raw file into an analysis-ready table

Dataset: 18 attributes

Customer: ID, age, gender, location

Purchase: item, category, amount (USD), size, color, season

Experience: review rating, shipping type

Behaviour: subscription, discount, promo code, previous purchases, payment method, purchase frequency

Python steps

- Loaded CSV with pandas and profiled shape, types and nulls
- Found 37 missing Review Ratings (the only gaps) and filled them
- Standardised column names to snake_case (e.g. purchase_amount)
- Engineered age_group to segment customers by age
- Loaded the clean table into PostgreSQL as "customer"

SQL analysis: 10 business questions

Queries run in PostgreSQL on the customer table

1 **Revenue by gender**
SUM, GROUP BY

2 **High-spending discount users**
Subquery

3 **Top 5 rated products**
AVG, ROUND, LIMIT

4 **Shipping vs spend**
Filtered AVG

5 **Subscriber economics**
Multi-metric GROUP BY

6 **Discount dependency**
CASE WHEN

7 **Customer segmentation**
CTE + CASE

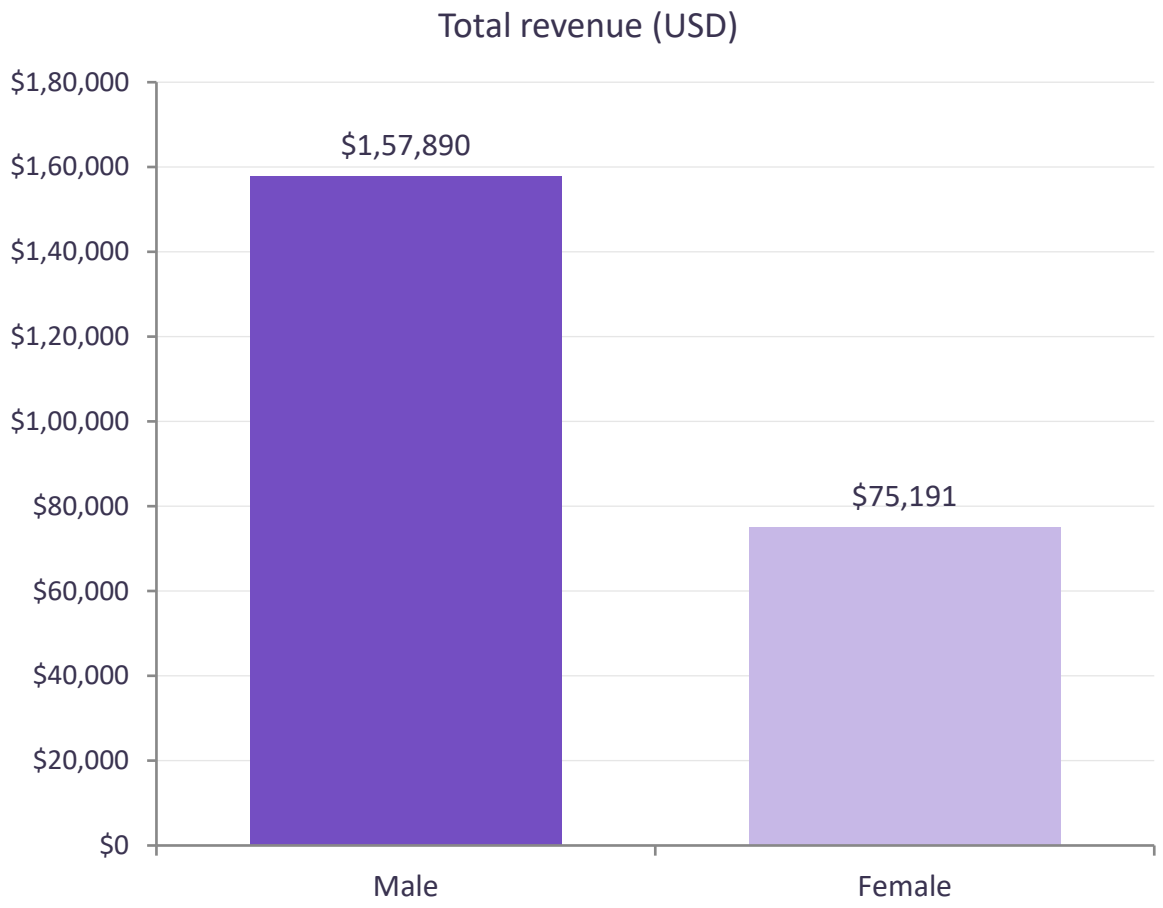
8 **Top 3 items per category**
ROW_NUMBER window

9 **Repeat buyers by plan**
Filtered COUNT

10 **Revenue by age group**
SUM, ORDER BY

Revenue analysis by gender

Query 1: males drive two-thirds of revenue through volume

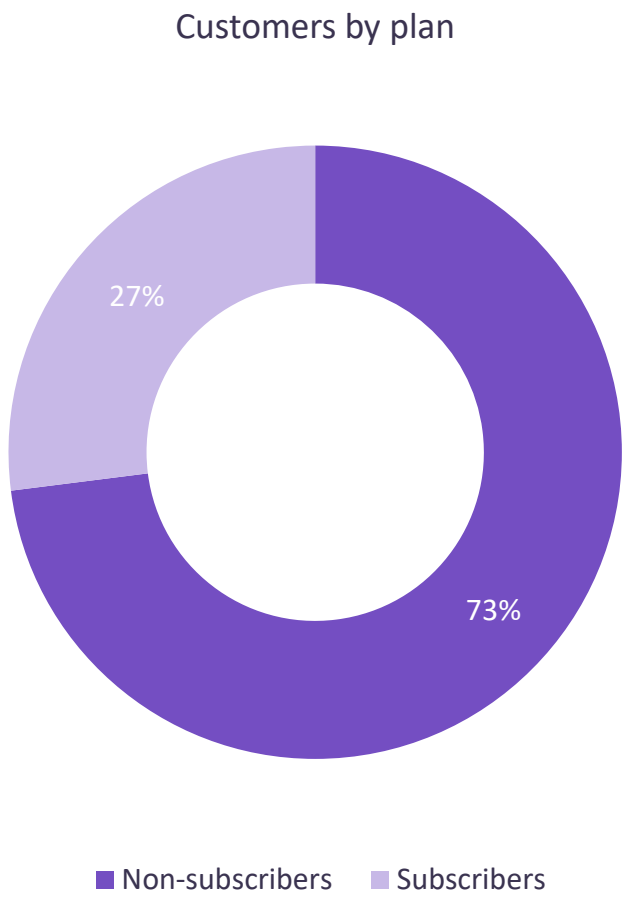


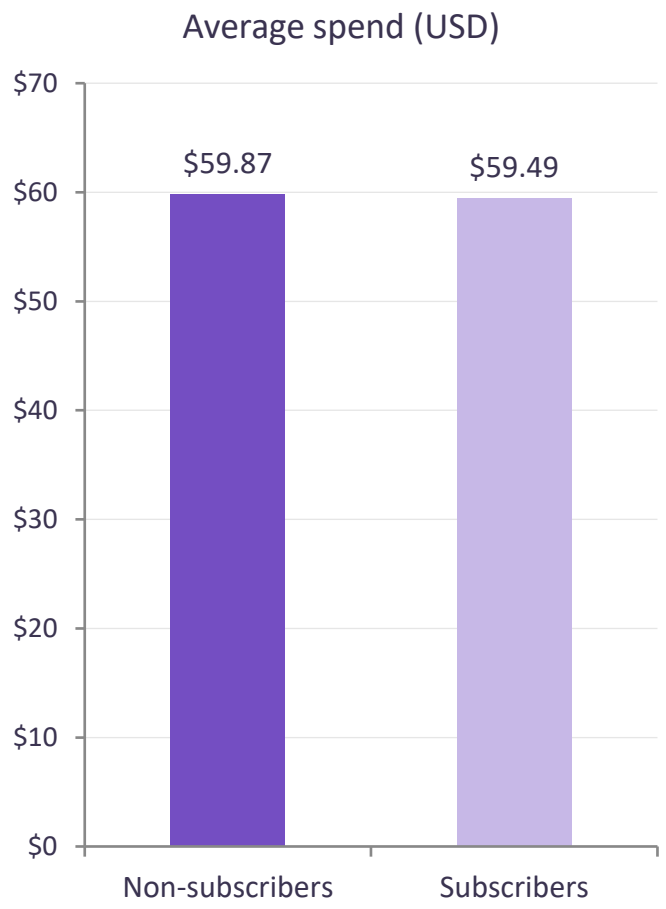
Insight

- Male: $157,890 from 2,652 customers (68%)
- Female: $75,191 from 1,248 customers (32%)
- Average spend is nearly equal ($59.54 vs $60.25), so the gap is customer count, not behaviour
- Opportunity: grow the female customer base, since each female customer is worth as much

Subscription and repeat-buyer analysis

Queries 5 and 9: subscribers are a small but equally valuable group



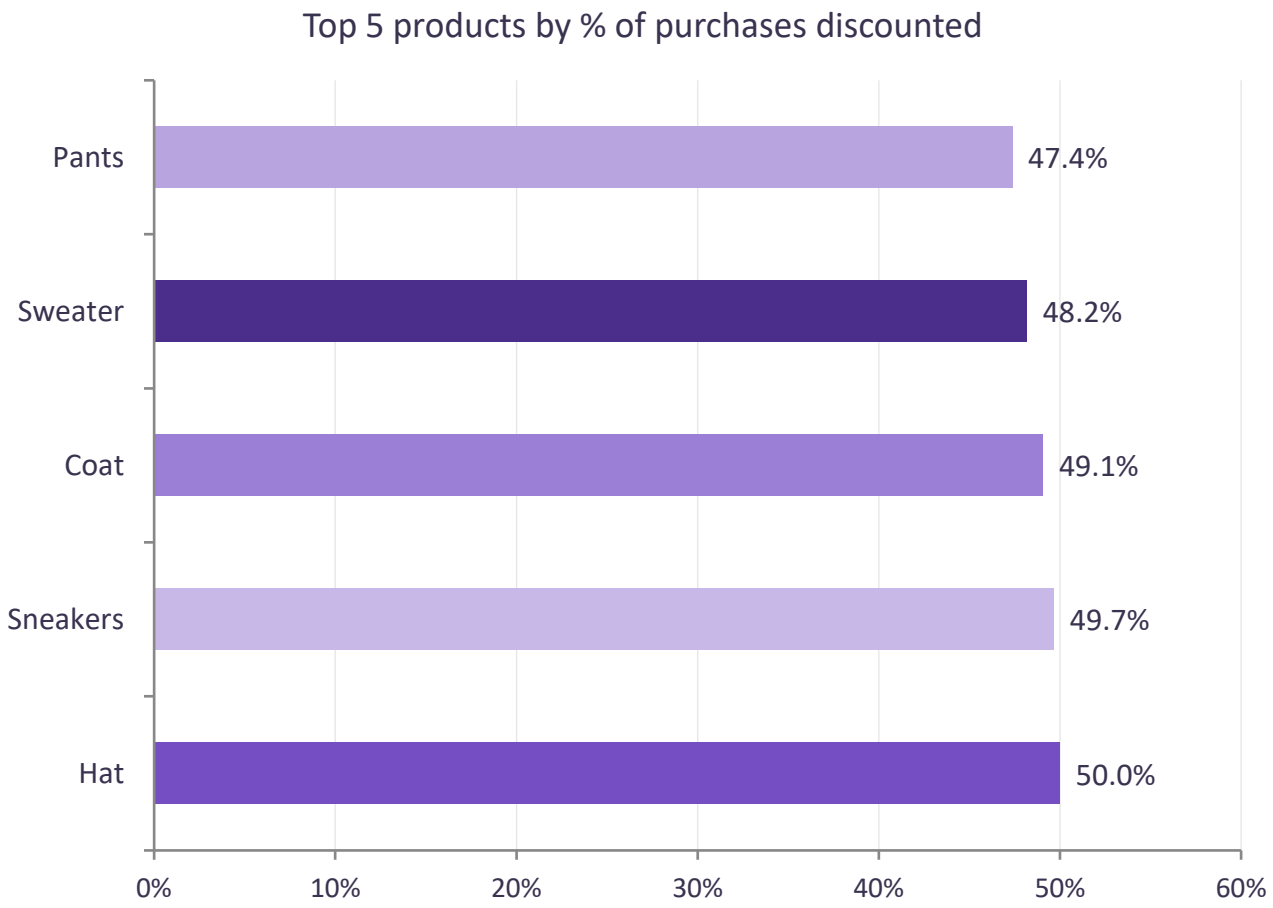


Insight

- Subscribers: 1,053 (27%), $62,645 revenue
- Non-subscribers: 2,847, $170,436 revenue
- Spend per head is identical ($59.49 vs $59.87)
- Of 3,476 repeat buyers (5+ prior purchases), only 958 subscribe: a big untapped base

Discount dependency

Queries 2 and 6: discounts are widespread but do not raise basket size

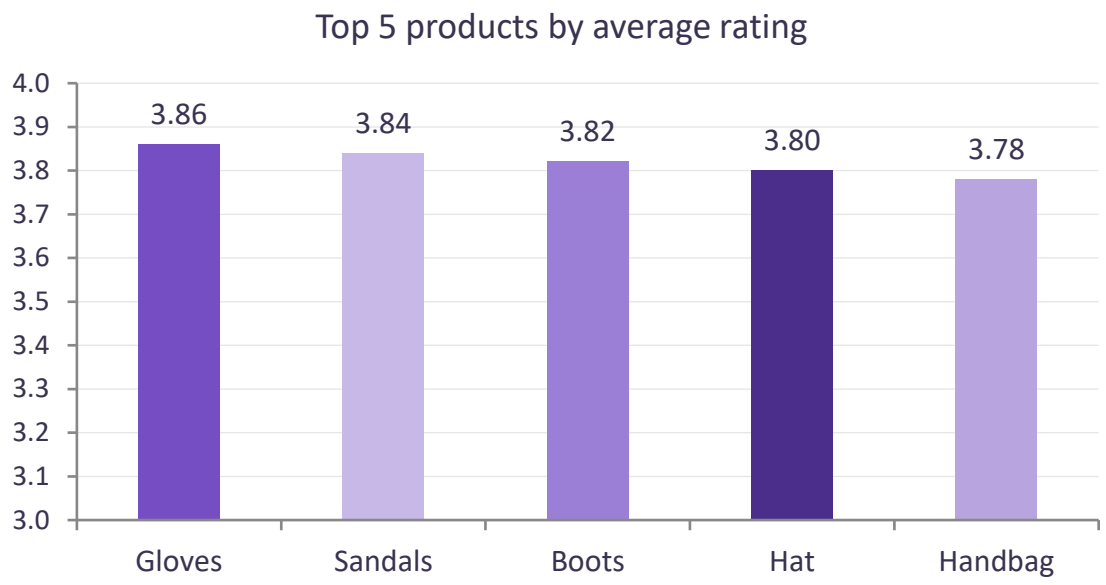


Insight

- 43% of all purchases carry a discount
- Half of all Hat purchases are discounted
- 839 discounted purchases were at or above the $59.76 average, so discounts are not only bargain hunting
- Avg spend: $59.28 with discount vs $60.13 without, so discounts give margin away without lifting basket value

Product rating and popularity

Queries 3 and 8: top-rated items and category bestsellers



Category	#1	#2	#3
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	n/a

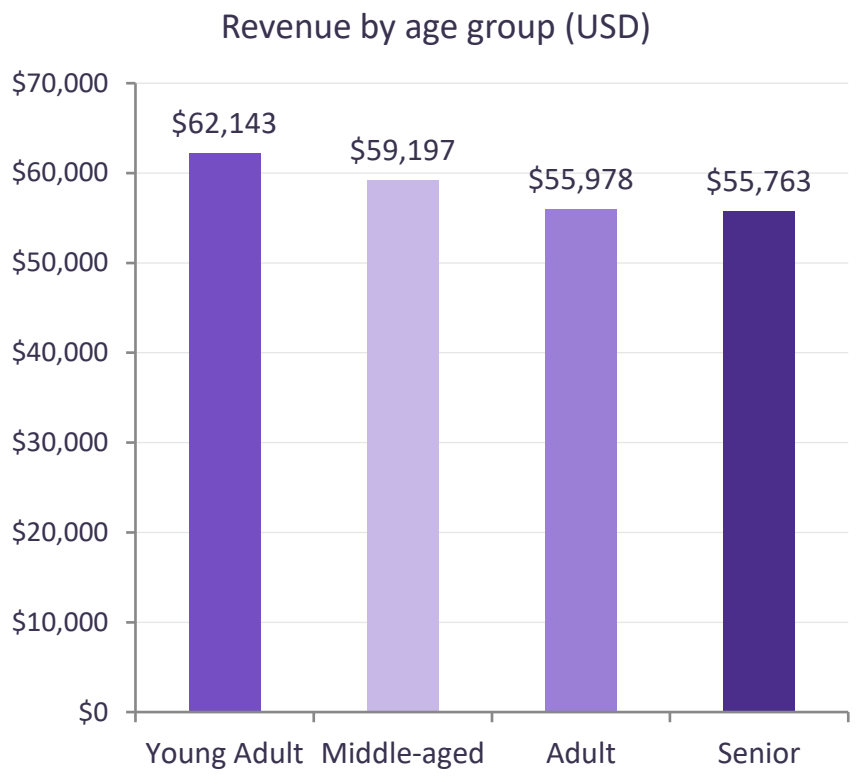
Insight

- Ratings are tightly clustered (3.78 to 3.86): no product is a clear quality outlier, and the 3.75 average leaves room to improve
- Order counts per item are very even (145 to 171), so merchandising should target margin and cross-sell, not just bestsellers

Shipping and age-group analysis

Queries 4 and 10: small differences, steady demand across ages





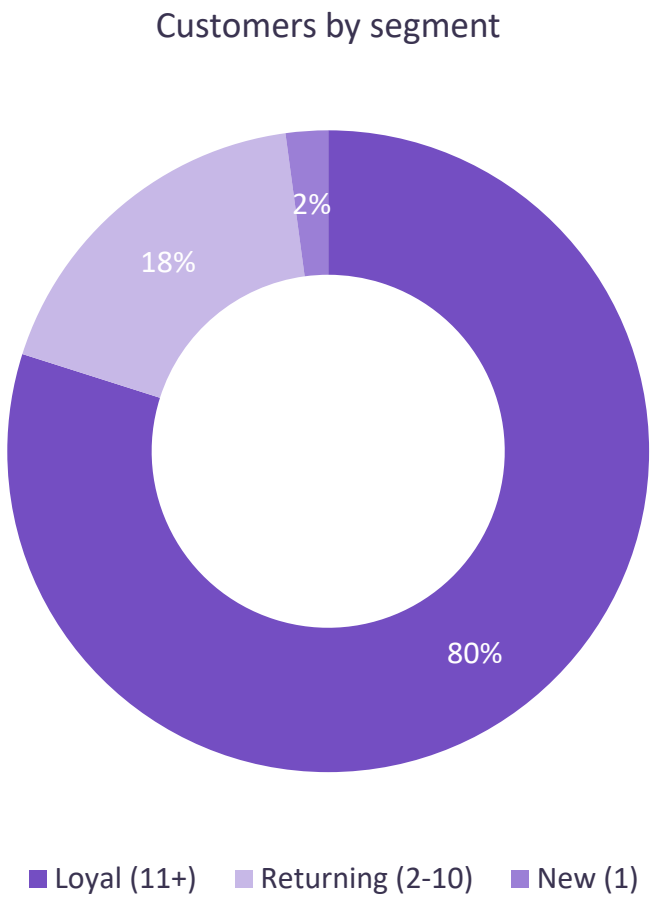
Insight

- Express beats Standard by $2.02 (3.5%)
- Young adults (18-31) lead at $62.1K
- Seniors trail at $55.8K, only 10% below the top

Age groups are quartile bands: 18-31, 32-44, 45-57, 58-70.

Customer segmentation

Query 7: a mature, loyal base with very few new customers



Insight

- Loyal: 3,116 customers (80%)
- Returning: 701 (18%)
- New: only 83 (2%)
- Retention is strong, but acquisition is thin: the base is not being refreshed
- Loyal customers are the natural target for subscription and referral programmes

Power BI dashboard

Interactive dashboard connected live to the PostgreSQL database



What it offers

- 4 KPI cards: customers, revenue, avg purchase, avg rating
- Slicers for gender, subscription, category, ship mode and payment mode
- 5 charts: subscription, gender, category, age band, top 10 products
- Clear-all button resets every filter

Key business insights

Six takeaways from the analysis

Volume, not basket size

Every segment averages about $60. Growing customer count is what grows revenue.

Clothing is the engine

$104K of $233K (45%) comes from Clothing; Accessories adds $74K (32%).

Subscription gap

Only 27% subscribe, though 80% of customers are loyal repeat buyers.

Discounts give margin away

43% of purchases are discounted with no lift in average spend.

Acquisition is the weak spot

New customers are only 2% of the base; female customers are 32%.

Quality is steady, not stellar

Ratings span 3.78 to 3.86 at the top; a 3.75 average has room to rise.

Recommendations

Turning insights into actions

1	**Launch a loyalty-linked subscription**	Offer perks (free shipping, early access) to the 3,000+ loyal non-subscribers.
2	**Make discounts targeted**	Cut blanket promos on Hats, Sneakers and Coats; reserve them for new or lapsing customers.
3	**Invest in acquisition**	Run campaigns for new and female shoppers to lift the 2% new-customer share.
4	**Double down on Clothing and Accessories**	Cross-sell Jewelry, Blouses and Pants, the top sellers, in bundles.
5	**Promote Express shipping**	Express carries a $2 higher basket than Standard; nudge it at checkout.
6	**Raise product quality scores**	Study low-rated items and replicate what drives Gloves and Sandals ratings.

Conclusion

A clean pipeline from Python to SQL to Power BI turned 3,900 transactions into clear direction: customers spend about $60 regardless of segment, so growth depends on acquiring more customers, converting loyal buyers to subscribers, and spending discounts more deliberately.

Tools used

Python (pandas, Jupyter)

PostgreSQL

Power BI